

The Eighth Information Systems International Conference (ISICO 2025)

RNN-based stock price forecasting through the integration of technical indicators, fundamental data, and news sentiment analysis

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Abstract

Accurately predicting stock prices is vital for making informed investment decisions in capital markets. Stock price volatility is influenced by a range of factors, such as global economic trends, monetary policy shifts, and overall market sentiment. This research proposes a stock price forecasting model utilizing deep learning techniques based on Recurrent Neural Networks (RNN). The model integrates three key types of stock price analysis: technical indicators, fundamental financial data, and sentiment analysis from financial news. The study focuses on three prominent Indonesian blue-chip stocks which are components of the LQ45 and IDX30 indices: Bank BCA (BBCA), Astra International (ASII), and Telkom Indonesia (TLKM). The used technical indicator includes moving averages and trading volume, while the fundamental analysis is obtained from annual financial ratios. Sentiment analysis is performed using IndoBERT on articles from the *detik.com* news portal. The combination of features from those three analyses enhances the accuracy of the stock price forecasting. Among the compared models, Bi-LSTM and Bi-GRU yielded the best results, with Bi-LSTM showing the lowest RMSE for TLKM (65.2) and ASII (119.49), while Bi-GRU showed the lowest RMSE for BBCA (97.79). The findings demonstrate the potential of integrated deep learning approaches to improve the precision and reliability of stock price prediction.

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Peer-review under responsibility of the scientific committee of the Eighth Information Systems International Conference (ISICO 2025).

Keywords: Stock price prediction; Recurrent Neural Network (RNN); technical analysis; fundamental analysis, news sentiment analysis.

1. Introduction

Accurate stock price forecasting plays a pivotal role in financial decision-making, both for individual investors and institutional stakeholders [1]. In capital markets, the ability to predict stock price movements effectively enables market participants to optimize portfolio allocation, manage risks, and identify opportunities for arbitrage and long-

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term gains [2]. Indonesia's capital market, administered by the Indonesia Stock Exchange (IDX), is a dynamic and growing segment of Southeast Asia's financial system. The overall performance of the Indonesian stock market is reflected by the Indeks Harga Saham Gabungan (IHSG) or Jakarta Composite Index (JCI). To classify market liquidity and performance, IDX also publishes various sub-indices, including LQ45 and IDX30, which consist of the most liquid and fundamentally strong companies, often referred to as blue-chip stocks. Stock price forecasting has traditionally relied on two main approaches: technical analysis and fundamental analysis [3]. Technical analysis involves studying historical price patterns and statistical indicators to forecast future price movements [4]. Fundamental analysis focuses on a company's intrinsic value by evaluating financial ratios derived from balance sheets and income statements [5]. However, while traditional approaches offer valuable insights, they often fall short in capturing the nonlinear, noisy, and highly dynamic nature of financial time series data.

As computers have become more powerful and more data is available, it has become easier to use machine learning and deep learning to predict stock prices. These techniques are capable of learning complex, nonlinear relationships from data, and they adapt over time as more data becomes available [6]. Various machine learning techniques have been applied to stock price prediction problems [7][8], including Support Vector Machines (SVM) [9], random forest [10], K-Nearest Neighbors (KNN) [11], artificial neural networks (ANN) [12], and also Recurrent Neural Networks (RNN) [13] with its variants such as Long Short-Term Memory (LSTM) [14] and Gated Recurrent Unit (GRU) [15]. Among these, RNN-based models, particularly LSTM and Bi-LSTM, have become popular due to their ability to capture long-term dependencies and avoid issues like vanishing gradients.

Although several studies have applied RNN-based models for stock price forecasting, many of them focus solely on technical indicators or historical price data. Some include fundamental analysis [16], but only a limited number incorporate sentiment analysis derived from financial news or public opinion, which are increasingly recognized as critical drivers of market behavior. Moreover, few studies have explored the integration of all three dimensions (technical, fundamental, and sentiment) in a unified deep learning framework, particularly within the context of the Indonesian stock market. Therefore, this research proposes an integrated stock price forecasting model based on RNN architecture, combining three key components, which are technical indicators extracted from historical trading data, fundamental financial metrics sourced from company financial reports, and sentiment scores derived from financial news articles. The methodology involves preprocessing multi-source data, constructing feature sets for each of the three input categories, and feeding them into RNN-based architectures. The models are trained and validated on stock data from Indonesian blue-chip companies, which are Bank Central Asia (BBCA), Astra International (ASII), and Telkom Indonesia (TLKM). This research aims to identify the best predictive model for each stock and assess the impact of integrating those three analyses on the prediction accuracy.

2. Research Methodology

This research proposed a stock price prediction model using Recurrent Neural Networks (RNN) variants, based on the integration of technical, fundamental, and sentiment analysis. The research methodology is illustrated in Fig. 1. The process begins with a literature review to explore existing methods, models, and findings related to stock price forecasting. The next process is described below.

2.1. Technical data processing

a) Data collection and preprocessing

Historical stock price data for BCA (BBCA), Astra (ASII), and Telkom (TLKM) is taken from Yahoo Finance. This data includes the closing price, the opening price, the highest price, the lowest, and the trading volume. All of this data is daily with a period from January 1, 2021 to December 31, 2023. Missing values are identified by checking whether there are null values in these columns. If missing values are found, they will be filled with the last available value before the value is null.

b) Analysis of technical indicators

The next step is to perform feature extraction on historical stock price data with the aim of obtaining new features, namely technical stock indicators. The technical indicators used are moving average, MACD (Eq. 1), and RSI (Eq. 2). The type of moving average used is SMA (Eq. 3) with a period or number of days for the average of 7

days, 14 days and 30 days. MACD (Eq. 1) uses the exponential moving average (EMA) with the period of 12 days (EMA_{12}) and 26 days (EMA_{26}). RSI (Eq. 2) uses the average gain divided by the average loss as the RS value. In SMA (Eq. 3), P_i is the closing price on the i -th period and n is the number of days (period).

$$MACD = EMA_{12} - EMA_{26} \quad (1)$$

$$RSI = 100 - \frac{100}{1 + RS} \quad (2)$$

$$SMA = \frac{\sum_{i=1}^n P_i}{n} \quad (3)$$

2.2. Sentiment data processing

a) Data collection and preprocessing

The financial news source is a news portal website, detik.com (www.detik.com). The process of scraping news data from Telkom, BCA, and Astra company begins by sending a request to detik.com, using specific keywords that represent each company to ensure that relevant news is taken. The example of the keywords used is ['bca', 'bbca', 'bank central asia'] for BBKA stock. After the web page is obtained, its HTML is extracted to obtain important information such as the date, title, and content of the article. Several processes are carried out, such as data cleaning and sentiment labeling. In the data cleaning process, each row of data is checked to ensure that all important columns (date, title, and news content) are filled in. If there is a null value in one or more of these columns, the data row is removed. The result of this process is 2500 clean news data of 3 companies that were analyzed.

Afterwards, we assign initial sentiment labels to 300 of the news data using a semi-supervised method with the help of TextBlob [17]. TextBlob is a Python library that can provide lexicon-based sentiment analysis based on patterns of words and phrases in the text. Each news article is translated into English and then analyzed using TextBlob to determine an initial sentiment score. This score ranges from -1 to 1, where 1 indicates positive sentiment, 0 indicates neutral sentiment, and -1 indicates negative sentiment. The sentiment score generated by TextBlob is added to the dataset as the data label of 300 news data.

b) Development and evaluation of sentiment analysis model

In this process, IndoBERT model training is carried out using the 300 news data that have been labeled using TextBlob. IndoBERT is a Transformer-based language model specifically designed for the Indonesian language [18]. The result of this process is an IndoBERT model that can analyze sentiment in news. Afterwards, the developed IndoBERT model for financial news sentiment analysis was evaluated using several metrics. The model was asked to predict the sentiment score of the labeled 300 news data and then the results were compared with the sentiment labels, which were obtained using TextBlob. The evaluation result of the IndoBERT model on predicting

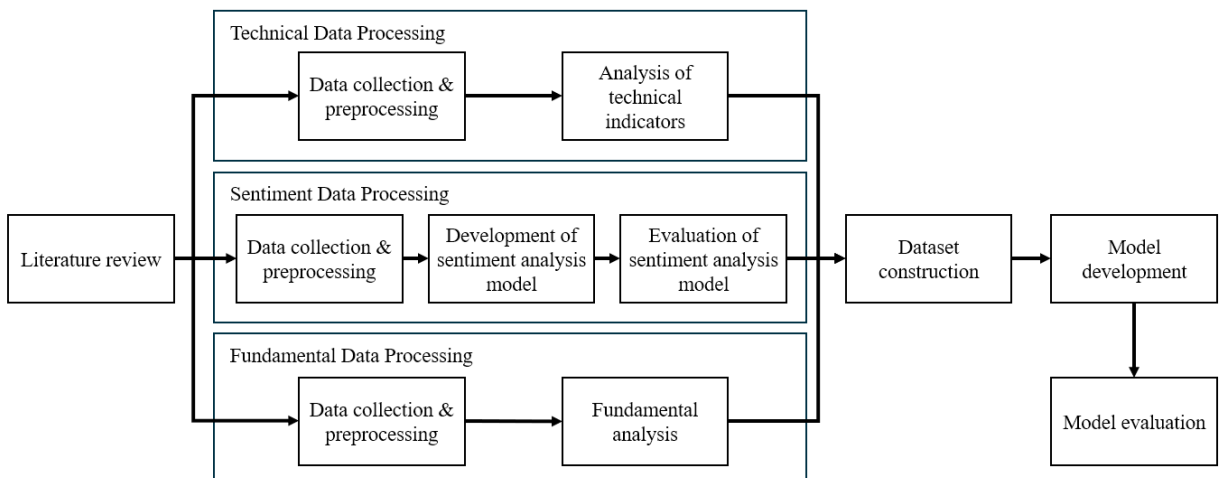


Fig. 1. Research methodology

the sentiment using 300 news data is the Mean Squared Error (MSE) of 0.018667, Root Mean Squared Error (RMSE) of 0.136629 and Mean Absolute Error (MAE) of 0.111002. Since the evaluation results are quite good, we use the developed IndoBERT model to label the sentiment score of the other 2200 news data. Therefore, each of the news article now has a sentiment score generated by the IndoBERT model. This score is then used as one of the feature inputs in the deep learning model used for stock price prediction.

2.3. Fundamental data processing

a) Data collection and preprocessing

The financial reports of BCA, Astra, and Telkom were obtained from the company's annual reports taken from investing.com for the period December 2019 to December 2023. The data taken are financials, balance sheets, and cashflow statements, specifically the Net Income, Ordinary Shares Number, Current Assets, Current Liabilities and Total Assets of each company.

b) Fundamental analysis

From the annual financial data, the value of the fundamental indicators to be used is then calculated, which are Earning Per Share (EPS), Return on Assets (ROA), and Current Ratio. EPS (Eq. 4) reflects the level of company profitability per share owned by investors. A high EPS indicates the company's ability to generate significant profits for shareholders. ROA (Eq. 5) measures the company's efficiency in managing its assets to generate net income. A high ROA is an indication that the company's assets are being utilized optimally. Current Ratio (Eq. 6) assesses the company's ability to meet its short-term obligations using available current assets. This ratio is the main indicator for measuring the company's liquidity. The fundamental data is still in the form of an annual period. Therefore, an interpolation process is done to convert annual fundamental data into daily data to align fundamental data with technical data and sentiment that are already in daily format.

$$EPS = \frac{\text{Net Income}}{\text{Shares Outstanding}} \quad (4)$$

$$ROA = \left(\frac{\text{Net Income}}{\text{Total Assets}} \right) * 100 \quad (5)$$

$$\text{Current Ratio} = \frac{\text{Current Asset}}{\text{Current Liabilities}} \quad (6)$$

2.4. Dataset construction

In this process, all data that has gone through preprocessing from various sources (historical, technical, sentiment, and fundamental data) is combined into one comprehensive dataset based on date. After the data is combined, on days that do not have sentiment data, the sentiment value will be made neutral or 0. Afterwards, feature normalization process is performed. All features in the dataset must be in a uniform value range so that the model can process the input better. Different features, such as volume, stock price, and financial ratios, often have very different value ranges. Therefore, min-max scaling is applied to the entire dataset so that all features have a scale between 0 and 1. The result of this process on TLKM stock is shown in Fig. 2.

2.5. Model development

The RNN model developed in this research is a multivariate model because the dataset used has more than 1 input feature. The RNN model requires data in the form of consecutive sequences to capture temporal patterns. Therefore, the data must be converted into an appropriate format, namely by creating input (X) and target (Y) pairs based on a certain time period called the look_back variable. The look_back variable determines the number of previous time steps used to predict the next time step. In this context, look_back is the number of previous days that will be used as input to predict the closing price of the stock on the next day. In this study, look_back is set to 10, which means the model will use data from the previous 10 days to predict the price on the eleventh day.

The RNN model is built using the Keras framework. This model consists of RNN layers. Each layer uses return_sequences=True to return the full sequence to the next layer, while the last layer returns the final output of the sequence. After that, the Dense layer is used to generate the stock price prediction output. The model

architecture used is shown in Fig. 3. The model input is divided into three different types of data paths, namely technical data, sentiment data, and fundamental data. The setting of parameters of this model is epoch=100, batch_size=16, optimizer=ADAM. We compared several RNN-based models, which are LSTM, GRU, Bi-LSTM, and Bi-GRU. Those models are selected due to their proven ability to model complex temporal dependencies in multivariate time series forecasting [19]. While LSTM and GRU address vanishing gradient issues, the bidirectional models (Bi-LSTM and Bi-GRU) were used to leverage both past and future information in sequences, which is particularly relevant for capturing sentiment dynamics [20].

2.6. Model evaluation

The model's performance is evaluated using the RMSE metric. The predictions generated by the model are converted back to their original scale through the inverse normalization process. This is done so that the resulting RMSE value is relevant and can be directly compared with the actual value of the stock price. The experiment was conducted to determine the influence of technical, sentiment, and fundamental features in predicting stock prices.

3. Results and Discussion

Based on the analysis of input data, the highest volatility of technical and fundamental data is found in ASII, and the lowest is BBKA. Meanwhile, in sentiment data, ASII has the most news data and generally has the most positive sentiment. The results from experiments on determining the influence of technical (T), fundamental (F), and sentiment (S) features for predicting stock price of TLKM, BBKA, and ASII measured using RMSE metric are shown in Table 1-3, respectively. Based on the results in Table 1, for TLKM stock price forecasting, it can be concluded that the combination of technical and sentiment features is the best combination of features in all RNN-based models, beating the combination of the three technical, fundamental, and sentiment features which produced the second-best performance. The best result is achieved using Bi-LSTM model with an RMSE score of IDR 65.2. In TLKM stock, the addition of fundamental features shows lower performance because TLKM's fundamental indicators have low values and high volatility. This shows that the performance of TLKM's fundamental indicators is not strong enough to predict stock prices in the short term. The Bi-LSTM's ability to capture long-range dependencies in both forward and backward time directions likely helps to better model such volatility [21]. However, the fundamental data is obtained by interpolating the annual data to daily frequency, which may cause

Date	Open	High	Low	Close	Volume	Current Ratio	EPS	ROA	MACD	RSI	SMA1	SMA2	SMA3	Sentiment
2023-12-13	3870.0	3920.0	3870.0	3920.0	54386800	0.778823	244.742521	8.475042	85.970569	75.806452	3918.571429	3830.000000	3690.333333	0.192151
2023-12-14	3880.0	3960.0	3880.0	3940.0	70705300	0.778807	244.844603	8.477721	86.353729	75.806452	3931.428571	3852.857143	3701.000000	0.000000
2023-12-15	3940.0	3980.0	3900.0	3980.0	109064200	0.778792	244.946685	8.480400	88.860722	75.000000	3940.000000	3874.285714	3714.666667	0.197723
2023-12-18	3940.0	3970.0	3930.0	3970.0	60940400	0.778746	245.252932	8.488438	89.014511	69.230769	3948.571429	3888.571429	3725.333333	0.168347

Fig. 2. The final dataset for TLKM stock price forecasting

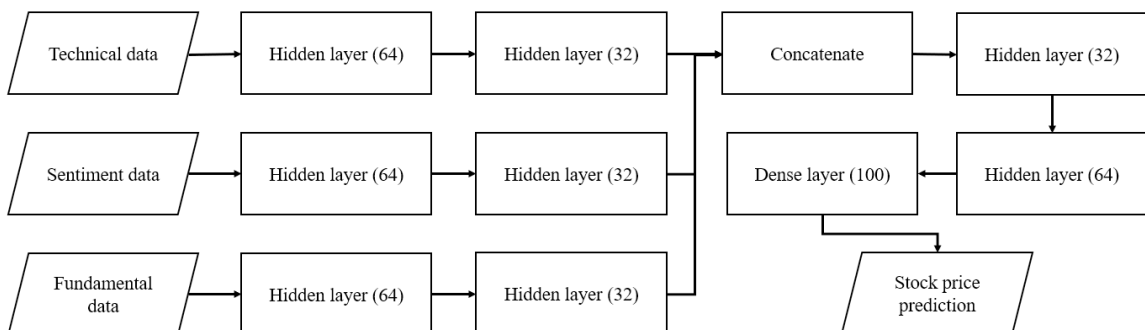


Fig. 3. The RNN model architecture

artificial smoothness, potentially underestimating abrupt financial shifts. This limitation may reduce the signal strength of fundamental features in the short-term prediction context, particularly in highly reactive stocks like TLKM.

Based on the results in Table 2, for BBKA stock price forecasting, the best combination of features varies for each model. The best result is achieved using the combination of technical and sentiment features using the Bi-GRU model with the RMSE score of IDR 97.79. Based on the correlation matrix analysis on BBKA stock price, the fundamental indicators ROA and EPS have a large influence, much greater than other stocks. For BBKA sentiment data, although the amount of news data is the least, it has the largest influence compared to other stocks, which shows that market reactions to the latest news and events are very important in predicting BBKA stock prices. The large influence of fundamental data and sentiment causes each model for BBKA stock forecasting to have a different best feature combination. The architecture and working mechanism of each model process data in different ways, so the most effective feature combination varies depending on how the model processes the input information. However, by averaging the RMSE of all models, the combination of the technical, fundamental, and sentiment features produces the lowest average RMSE. In this case, Bi-GRU, which balances modeling depth with computational efficiency, can better capture BBKA's sharp yet structured responses to news and earnings performance [22].

Based on the results in Table 3, for ASII stock price forecasting, the combination of the technical, fundamental, and sentiment features gives the best results in all types of models. The best result is achieved using the Bi-LSTM with the RMSE score of IDR 119.49. ASII has the best fundamental and sentiment indicator values compared to other stocks, but based on correlation analysis, the influence of both features is still relatively small compared to the others. Therefore, ASII stock price forecasting requires a combination of the three features to be able to obtain optimal results. The bidirectional memory of Bi-LSTM allows the model to extract subtle signals that may not be immediately evident from simple correlation. These findings indicate that the model's performance is highly context-dependent, and understanding the underlying data characteristics is essential for selecting the appropriate architecture.

The visualization of the proposed stock price forecasting is shown in Fig. 4, in which the blue line represents the training data, the orange line represents the actual value of the test data, and the green line represents the forecasting result. The proposed RNN-based model can predict price changes when there is news that affects price movements. For example, in TLKM stock, the merger of Indihome with Telkomsel, which was officially announced on July 1, 2023, caused a drastic drop in TLKM share prices from IDR 4000 at the end of June 2023 to IDR 3720 at the end of July 2023, as shown in Fig. 4a. For BBKA stock, in October 2023, there was an increase in interest rates in the

Table 1. Influence of the Input Features Combination on TLKM Stock Price Forecasting (IDR)

Method	T	F	S	T+F	T+S	F+S	T+F+S
LSTM	86.48	99.7	93.13	79.45	75.25	88.15	79.93
GRU	84.23	90.92	89.68	88.12	75.2	91.44	79.27
Bi-LSTM	96.51	104.86	109.99	94.1	65.2	97.28	85.06
Bi-GRU	83.18	93.78	108.6	89.34	68.03	118.8	75.3

Table 2. Influence of the Input Features Combination on BBKA Stock Price Forecasting (IDR)

Method	T	F	S	T+F	T+S	F+S	T+F+S
LSTM	128.89	126.76	119.81	124.56	138.03	124.46	112.27
GRU	111.6	110.29	102.55	125.5	123.22	104.55	110.43
Bi-LSTM	118.15	120	129.34	108.52	120.92	118.6	121.63
Bi-GRU	101.93	113.02	118.14	101.15	97.79	126.92	104.79

Table 3. Influence of the Input Features Combination on ASII Stock Price Forecasting (IDR)

Method	T	F	S	T+F	T+S	F+S	T+F+S
LSTM	140.87	138.64	143.21	148.54	150.97	154.8	121.92
GRU	143.05	153.13	147.39	145.87	143.38	154.29	120.86
Bi-LSTM	150.29	187.34	166.16	156.14	149.25	213.58	119.49
Bi-GRU	149.34	170.87	149.63	161.65	133.17	151.52	123.91

United States, causing the IHSG to fall. At the same time, Bank Indonesia's interest rate also rose to 6%. In addition, there was also a weakening of the rupiah against the dollar to almost 16,000. All of this caused the price of BBKA shares on October 16, 2023, from a price of IDR 9100, to fall to IDR 8600 on November 1, 2023, as shown in Fig. 4b. In ASII stock, there is news about Daihatsu's crash test manipulation. On April 28, 2023, Daihatsu admitted to manipulating the side crash test certification procedure on several of their car models. This impacted the ASII stock price, so that it dropped drastically from IDR 6750 on April 28, 2023, to continue dropping to IDR 6125 on May 8, 2023. The model was able to predict this event based on the news sentiment, but was a little late, especially on May 5, 2023, because it was indeed a very drastic price drop (Fig. 4c).

4. Conclusion

This research proposed a stock price forecasting model by integrating technical indicators, fundamental financial data, and news-based sentiment analysis using an RNN-based method. The method was implemented to predict the price of several blue-chip stocks, such as TLKM, BBKA, and ASII. The dataset is taken from Yahoo Finance and investing.com (January 1, 2020 - December 3, 2023) to obtain the technical indicators (SMA, MACD, and RSI) and fundamental indicators (Current Ratio, ROA, and EPS). Sentiment analysis is done by scraping related news from detik.com, and the sentiment score is calculated using IndoBERT. A comparison of LSTM, GRU, Bi-LSTM, and Bi-GRU models was conducted to determine the best combination of features for each stock. The experimental results showed that the combination of the three stock analysis features did not always produce the best performance for each stock. For TLKM, all models performed best in the combination of technical features and news sentiment, with the best model being Bi-LSTM with an RMSE value of IDR 65.2. This is because TLKM does not have a strong fundamental value. For BBKA shares, the combination of each feature is different for each model, although if averaged, the combination of the three features gives the best performance. The best performance is given by Bi-GRU using a combination of technical features and sentiment with an RMSE of IDR 97.79. For ASII shares, the combination of the three features gives the best performance for all models, with the Bi-LSTM model giving the lowest RMSE of IDR 119.49. Based on these results, the combination of technical, fundamental, and news sentiment indicators can generally increase the accuracy of stock closing price predictions.

This study focuses on three liquid blue-chip stocks to ensure consistent data availability. However, the forecasting of mid-cap or sector-diverse stocks can be considered for future research to broaden the generalization. To further improve the predictive performance, several strategies can be employed, such as the development of hybrid models that combine the strengths of various deep learning architectures. Additionally, incorporating an attention mechanism could further improve performance by allowing the model to focus on the most relevant

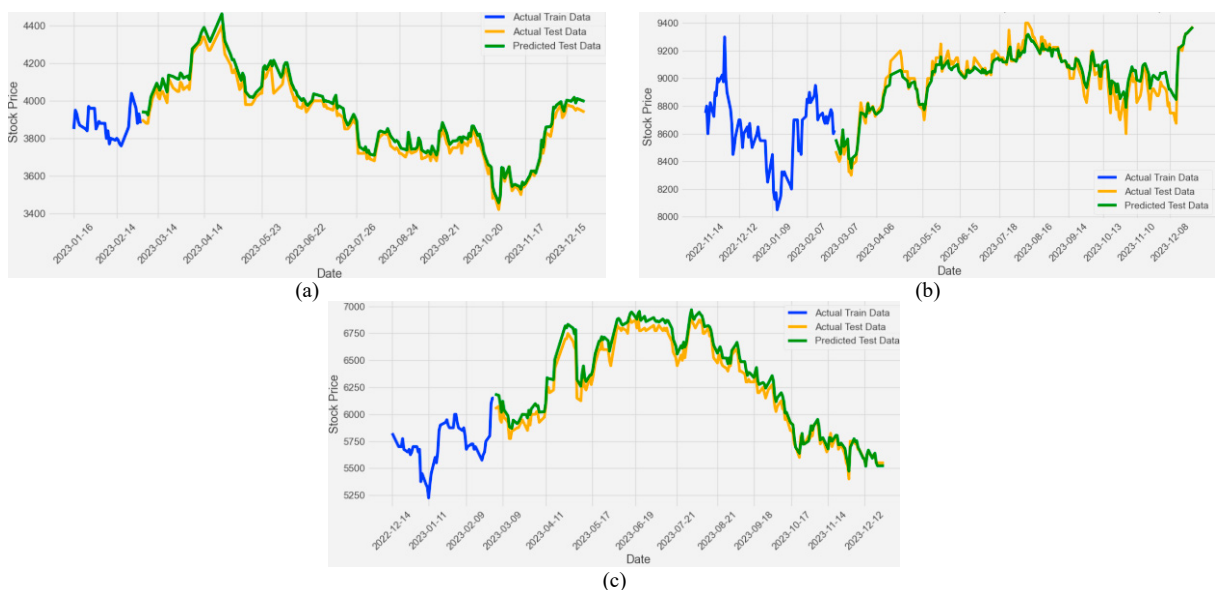


Fig. 4. The visualization of stock price forecasting on (a) TLKM; (b) BBKA; and (c) ASII

features during prediction. The inclusion of additional technical and fundamental indicators can also be considered to enrich the feature set and potentially boost the model's predictive accuracy.

Acknowledgements

This research is supported by Institut Teknologi Sepuluh Nopember (ITS) through the Departmental Research scheme with grant number 2798/PKS/ITS/2025.

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